

Optional: Gamestop stock vs Tesla

Determining the price of a stock is complex. It depends on several factors, including the number of outstanding shares and expectations about the company's future profits.

One key factor is **profit growth**. If a company's profits are increasing, its stock price will often rise. In such cases, investors may choose to **buy the stock** in anticipation of future gains.

But what if you expect the stock price to **decrease**?

This is where **short selling** comes in. Short selling is a strategy used to profit from a decline in a stock's price. In this process, an investor:

1. Borrows shares of a stock
2. Sells them at the current market price
3. Buys them back later at a lower price
4. Returns the shares to the lender

If the price drops, the investor makes a profit. However, if the price rises instead, the investor can incur significant losses.

Short selling is typically used by experienced investors, such as hedge funds. While stock prices often fall faster than they rise, making quick profits possible, the risks can be high.

Real-World Examples

Short sellers do not always predict correctly. For example, many investors once bet against **Tesla**, expecting its stock price to fall. However, as Tesla became more profitable, its stock price increased, leading to losses for those who had shorted it.

In some cases, stock prices rise due to factors beyond company performance. A well-known example is **GameStop**, a struggling video game retailer. Individual investors on the Reddit forum **WallStreetBets** began buying large amounts of GameStop stock, driving its price sharply upward. This caused significant losses for hedge funds that had shorted the stock.

Your Task

As a data scientist working for a hedge fund, you will:

- Extract profit data for companies like **Tesla** and **GameStop**
- Analyze stock price trends
- Build a dashboard comparing **stock prices and company profits**